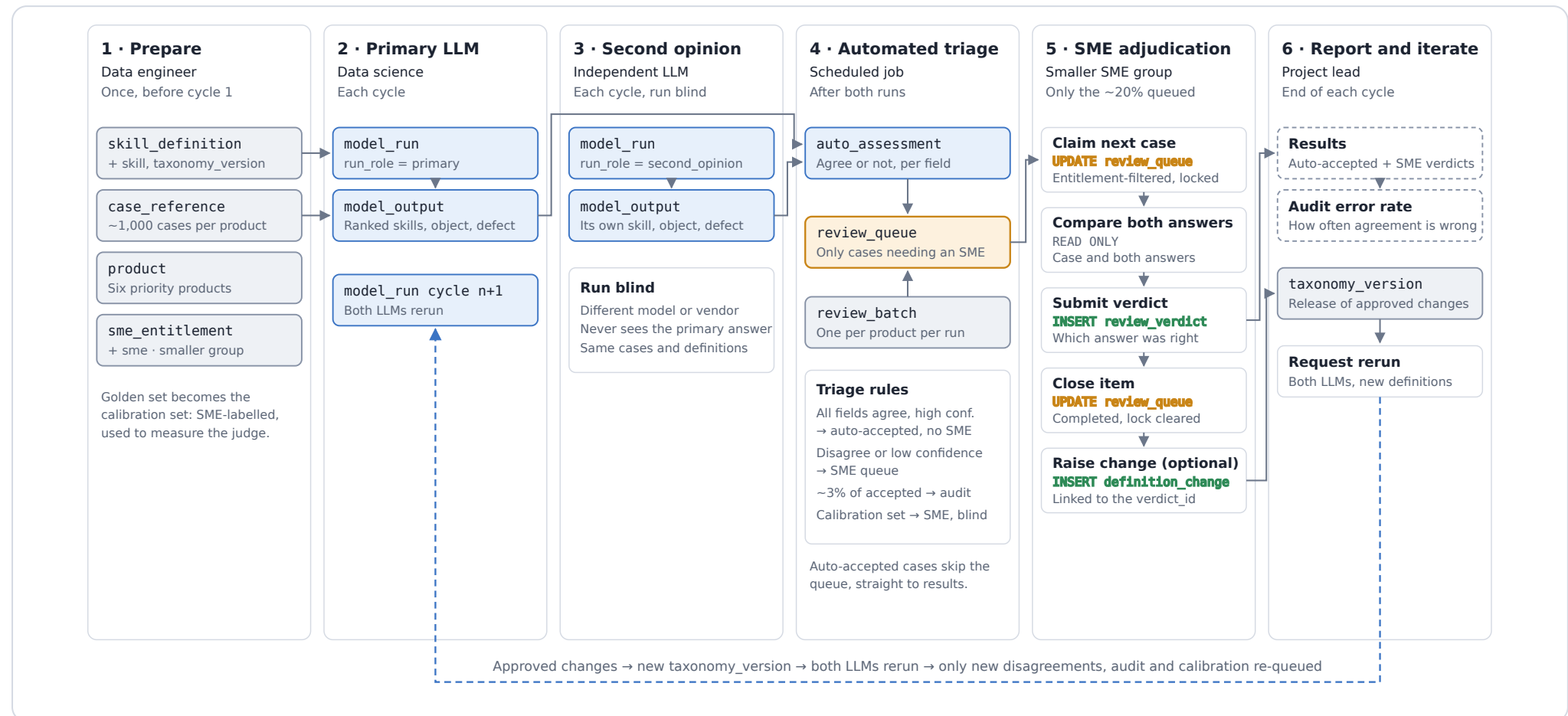


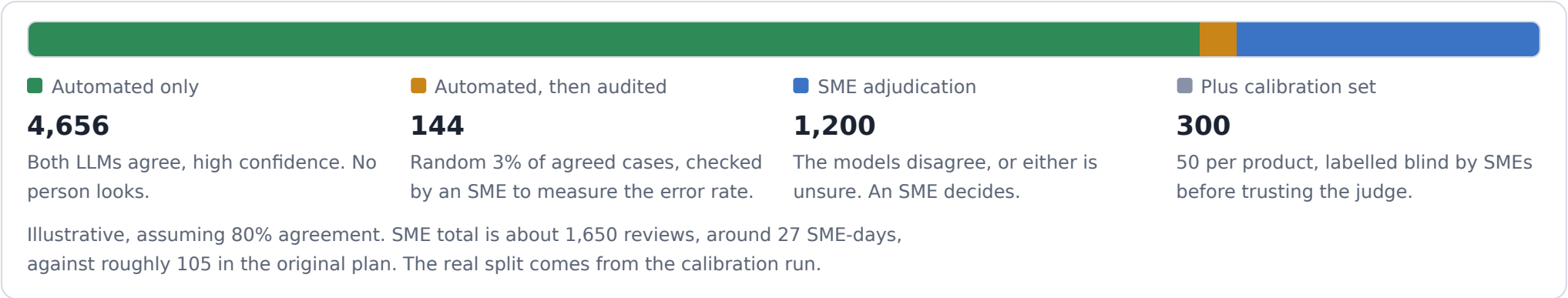
Validation Data Flow

Revised plan: a second, independent LLM checks every case, and agreement is accepted automatically. SMEs only review what the two models disagree on, plus an audit sample of what they agreed on. Around 80% of cases never reach a person.

- Loaded once, fixed
- Written once per run, never changed
- Updated in place (queue only)
- Append only, rows never edited
- Derived or reported, not stored



Where 6,000 cases go in cycle 1



What moves along each arrow, and when

FROM → TO	WHAT MOVES	WHEN
Data engineer → both LLMs	The same case numbers and the same definition text for the current taxonomy_version	Before each cycle
Primary LLM → model_output	Ranked skills, object, defect, explanation and confidence under a run_id with run_role = primary	After each run
Second LLM → model_output	Its own answer, produced without seeing the primary one, under a second run_id with run_role = second_opinion	After each run
Triage job → auto_assessment	Per case: skill agrees, object agrees, defect agrees, lowest confidence, outcome and the rule version that decided it	Once both runs are in
Triage job → review_batch, review_queue	Only disagreements, low confidence, audit sample and calibration cases. Agreed cases are never queued	After triage, before batches open
Form → review_queue	Claim, complete, skip or escalate. The only table the form updates	During review
Form → review_verdict, definition_change	New rows only, including which answer the SME sided with	On submit
auto_assessment + review_verdict → results	Accepted cases take the agreed answer. Queued cases take the SME's. The audit sample gives the error rate on the accepted ones	End of each cycle
definition_change → taxonomy_version → rerun	Approved wording goes to both LLMs	Between cycles

Schema changes from the previous design

TABLE	CHANGE	WHY
model_run	Add run_role (primary, second_opinion) and paired_run_id	The second LLM's output has the same shape, so it reuses model_output rather than a new table
auto_assessment NEW	One row per case per run pair: field agreements, confidences, outcome, rule_version	A stored record of why each case was or wasn't sent to a person. Thresholds will be tuned, so the decision must be traceable
review_queue	New routing reasons: llm disagreement, low confidence, audit sample, calibration	Reports can split the SME effort by why each case was sent
review_verdict	Add sided_with (primary, second_opinion, both, neither)	Shows which model is more often right when they disagree
ml_model_run, ml_prediction	Retire, or keep as an optional third signal	The second LLM replaces the classifier's job

What the plan depends on

Agreement isn't correctness

Two LLMs reading the same definition can make the same mistake, especially where the wording is ambiguous. The audit sample is the only thing that measures this, so it shouldn't be cut to save time.

The judge must be calibrated first

Run the 300 calibration cases before trusting auto-acceptance. If the second LLM is wrong often on cases SMEs labelled, raise the confidence threshold before going to full volume.

Object and defect are free text

"Controller B" and "storage controller B" won't match exactly. Compare the skill code directly, and use an LLM equivalence check or a controlled list for object and defect.

80% is a setting, not a result

The confidence threshold decides how much is auto-accepted. Tune it on the calibration set until the audit error rate is acceptable, and report the resulting share honestly.

The form's contract

Reads

- review_queue, review_batch, sme_entitlement to pick the next case
- case_reference for the case detail
- model_output for both runs, primary and second opinion, shown side by side
- auto_assessment to highlight the fields that disagree
- skill, skill_definition at the run's taxonomy version

Updates

- review_queue only: status, assigned SME and lock time

Calibration and audit cases are shown blind. Neither model's answer appears until the SME has entered their own.

Inserts

- review_verdict: one row per completed review, including sided_with
- definition_change: optional, linked to the verdict

Never writes

case_reference, model_run, model_output, auto_assessment, skill_definition and review_batch. If a design needs one of these written from the review screen, raise it with the data team first.

The triage job, the audit sampling and the stale-lock reset are the data team's to build. The form only claims, completes, skips and escalates rows that already exist.